COMPUTER NETWORKS PROJECT VIVA PREPARATION

1. OVERALL NETWORK ARCHITECTURE

Our project implements a university network consisting of multiple departments connected through a centralized core infrastructure.

Departments:

- Admission (VLAN 10)
- Exam (VLAN 20)
- Admin (VLAN 40)
- Finance (VLAN 50)
- Marketing (VLAN 60)
- Server Farm (VLAN 70)

All departments connect to a Core Switch (CS-01), which connects to Router R-01. Inter-VLAN communication is achieved using Router-on-a-Stick.

2. VLANS

A VLAN (Virtual Local Area Network) logically separates devices into different broadcast domains even when they are connected to the same physical switch.

This is a fancy way of saying theres a single switch on the physical layer, but in data linking layer which is layer 2 we are basically making a virtual switch for each department i.e vlan 10 for admission is like its own separate virtual switch even through physically all vlans lie on the same core switch which connects to each department in our project, im trying to explain it but its hard to understand without visualizing, kindly go to chatgpt or claude for better explination or visual help.

Benefits:

- Better security
- Better organization
- Reduced broadcasts
- Easier management

Each department was placed in its own VLAN to separate traffic and improve security and network organization.

3. VLAN ASSIGNMENTS

Admission → VLAN 10
Exam → VLAN 20
Admin → VLAN 40
Finance → VLAN 50
Marketing → VLAN 60
Server Farm → VLAN 70

4. TRUNK PORTS

A trunk port carries traffic for multiple VLANs over a single physical link.

Without trunks, we would need a separate cable for every VLAN. Trunks allow all VLAN traffic to travel between switches and the router through a single connection.

Examples:

- CS-01 ↔ SW-ADM
- CS-01 ↔ SW-EXAM
- CS-01 ↔ SW-MKT

5. ACCESS PORTS

An access port belongs to only one VLAN and is typically connected to end devices such as PCs, servers, and laptops.

Example:
Admission PCs belong to VLAN 10.

6. ROUTER-ON-A-STICK

A method that allows one physical router interface to route traffic for multiple VLANs using subinterfaces.

Its like each router gigaehternet port further divides itself into more virtual subinterfaces, originally it was like gigaethernet 0/0 interface, then it divides into gigaethernet0/1.10 subinterface and so on.

Example:
G0/1.10 → VLAN 10
G0/1.20 → VLAN 20
G0/1.40 → VLAN 40
G0/1.50 → VLAN 50

G0/1.60 → VLAN 60
G0/1.70 → VLAN 70

To enable communication between different VLANs while using a single router interface.

7. DHCP

Dhcp basically dynamically assigns ips to pcs from a range of ips that we define in its ip pool, pool is like a range of available ips.
Dynamic Host Configuration Protocol automatically assigns:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server

It eliminates manual IP configuration and reduces configuration errors.

8. CENTRALIZED DHCP

Centralized DHCP is easier to manage than maintaining separate DHCP servers in every department.

DHCP Server:
SRV-01 (10.10.7.10)

9. DHCP RELAY

DHCP broadcasts cannot cross routers. DHCP Relay forwards client DHCP requests to a remote DHCP server.

Command used:
ip helper-address 10.10.7.10

The DHCP server is in VLAN 70 while clients are in other VLANs.

10. DNS

Domain Name System converts names into IP addresses.

Example:
srv01.company.local

→
10.10.7.10

Humans remember names easier than IP addresses.

11. STATIC Ips

Critical services must always be reachable at predictable addresses.

Examples:
SRV-01 → 10.10.7.10
SRV-02 → 10.10.7.11
SRV-03 → 10.10.7.12
SRV-04 → 10.10.7.13

12. ACCESS POINT

To support wireless users while keeping them inside VLAN 60.

They receive DHCP addresses from the centralized DHCP server through the same VLAN.

13. SERVER FARM

Centralized servers are easier to secure, manage, and maintain.

Server Farm:
VLAN 70
Subnet:
10.10.7.0/24

14. ISP ROUTER

It represents the organization's connection to the Internet and future external connectivity.

15. MOST IMPORTANT VIVA QUESTION

Admin and Finance devices were initially left in VLAN 1 instead of VLAN 40 and VLAN 50. As a result, DHCP requests failed. Creating the correct VLANs and assigning the access ports resolved the issue.

This answer demonstrates practical troubleshooting experience.